

WEBVTT

110

00:25:45.540 --> 00:25:46.560

Isaiah Osborne: Hey! Hurry up, and.

111

00:25:49.250 --> 00:25:50.998

Karson Woods: Not too bad. How about yourself?

112

00:25:51.430 --> 00:25:52.490

Isaiah Osborne: Doing well.

113

00:25:54.830 --> 00:25:57.100

Richard Hoehn: Or perfect school here, or something.

114

00:25:58.840 --> 00:26:00.512

Isaiah Osborne: Definitely excited for break.

115

00:26:00.930 --> 00:26:02.600

Karson Woods: Oh, yeah, for sure.

116

00:26:04.930 --> 00:26:11.270

Karson Woods: Others looking better. And I just checked recent. So hopefully, that holds out, oh, yeah, definitely.

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00:26:11.970 --> 00:26:17.719

Richard Hoehn: Somebody doesn't technically follow their children actually just following branch, or.

118

00:26:18.060 --> 00:26:25.500

Richard Hoehn: which is which is like, why.

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00:26:25.630 --> 00:26:27.900

Richard Hoehn: let's say that little question here.

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00:26:28.140 --> 00:26:29.010

Richard Hoehn: Well.

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00:26:30.180 --> 00:26:32.699

Isaiah Osborne: Richard, if you're talking, we can't hear what you're saying.

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00:26:33.168 --> 00:26:39.399

Richard Hoehn: Just follow your kids, because if they're that young they might bounce around and they're just gonna follow them.

123

00:26:39.520 --> 00:26:41.980

Karson Woods: He's probably not his computer right now.

124

00:26:42.570 --> 00:26:44.330

Isaiah Osborne: Yeah, it's kind of what it sounds like.

125

00:26:49.180 --> 00:26:50.020

Richard Hoehn: Try it.

126

00:26:51.100 --> 00:27:02.950

Richard Hoehn: The truth is probably just come over here, but call out.

127

00:27:03.870 --> 00:27:04.809

Isaiah Osborne: You doing? Hector?

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00:27:05.350 --> 00:27:07.010

Hector Rogel Jr.: Doing? Well, how about yourself?

129

00:27:07.240 --> 00:27:09.240

Isaiah Osborne: Good, excited for break.

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00:27:09.690 --> 00:27:10.530

Hector Rogel Jr.: Yes.

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00:27:11.228 --> 00:27:14.709

Isaiah Osborne: You doing anything fun, or you just staying around here.

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00:27:16.130 --> 00:27:17.349

Hector Rogel Jr.: Staying around here.

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00:27:17.500 --> 00:27:18.569

Isaiah Osborne: Yeah, I hear ya.

134

00:27:19.230 --> 00:27:20.369

Hector Rogel Jr.: It's just gonna work.

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00:27:22.284 --> 00:27:23.009

Isaiah Osborne: Yes.

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00:27:23.010 --> 00:27:27.730

Richard Hoehn: We don't know what's special about it.

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00:27:27.880 --> 00:27:38.260

Richard Hoehn: Interesting? Because I think you'll be like, oh, this is actually a special watch. It's not.

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00:27:39.280 --> 00:27:48.230

Richard Hoehn: It's only so only green corpse.

139

00:27:48.580 --> 00:27:51.220

Karson Woods: What the heck guys had a chance to look at the notebook yet.

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00:27:52.796 --> 00:27:56.230

Isaiah Osborne: I've downloaded it, but I haven't actually tried to run it yet.

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00:27:56.400 --> 00:27:57.320

Karson Woods: Okay.

142

00:27:58.120 --> 00:27:59.960

Karson Woods: Yeah.

143

00:28:00.740 --> 00:28:03.380

Isaiah Osborne: I was going to ask if you've looked at it, and I.

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00:28:03.570 --> 00:28:06.830

Karson Woods: No, I haven't. I'm about to look at it right now. Actually.

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00:28:06.830 --> 00:28:07.520

Isaiah Osborne: Okay.

146

00:28:07.760 --> 00:28:12.550

Hector Rogel Jr.: I looked at it like about 20 min ago, and and it's a lot.

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00:28:12.550 --> 00:28:13.183

Karson Woods: Is it.

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00:28:14.210 --> 00:28:15.310

Hector Rogel Jr.: I think it's a.

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00:28:15.910 --> 00:28:16.730

Isaiah Osborne: That's great!

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00:28:17.090 --> 00:28:18.109

Karson Woods: That's that's nice.

151

00:28:18.110 --> 00:28:19.070

Richard Hoehn: Hey, guys.

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00:28:19.720 --> 00:28:20.620

Isaiah Osborne: Hey! How are you doing.

153

00:28:22.020 --> 00:28:25.079

Richard Hoehn: Let me make sure this is on. Okay.

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00:28:35.970 --> 00:28:37.855

Richard Hoehn: I just made sure the meeting.

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00:28:39.230 --> 00:28:43.229

Richard Hoehn: the AI companion thing was on, how are you guys doing.

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00:28:43.620 --> 00:28:44.690

Isaiah Osborne: Doing well.

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00:28:46.720 --> 00:28:48.120

Richard Hoehn: Good good

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00:28:57.960 --> 00:29:01.759

Richard Hoehn: So I only have about half an hour. Maximum kind of thing.

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00:29:03.380 --> 00:29:17.889

Richard Hoehn: do you guys want to maybe look over what our deliverables are, and all that, and just kind of think about what we need to do or or do. Do you know whereas Isaiah, have you already read it. You could just tell us kind of thing or.

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00:29:18.651 --> 00:29:22.460

Isaiah Osborne: I've looked at it a little bit. I have.

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00:29:22.920 --> 00:29:25.569

Isaiah Osborne: I'll be honest. I have not looked at this one as much.

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00:29:25.790 --> 00:29:26.740

Richard Hoehn: Okay.

163

00:29:27.730 --> 00:29:31.180

Isaiah Osborne: But hold on, I'm looking at it right now.

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00:29:31.730 --> 00:29:37.599

Isaiah Osborne: So our main deliverable is going to be. We have our Github repo as normal.

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00:29:37.840 --> 00:29:39.200

Richard Hoehn: Good job. Thank you.

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00:29:39.735 --> 00:29:43.640

Isaiah Osborne: And then we have a final report that's due.

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00:29:44.650 --> 00:29:49.799

Isaiah Osborne: So in that report, we're gonna have to look at bias in

the data itself.

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00:29:50.600 --> 00:29:53.899

Isaiah Osborne: Any algorithmic bias we can find

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00:29:55.120 --> 00:30:01.400

Isaiah Osborne: and then we have to download data for Ramsey County

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00:30:02.110 --> 00:30:05.579

Isaiah Osborne: instead of the original Hinnepin county data that he gives us.

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00:30:07.190 --> 00:30:10.340

Isaiah Osborne: And then kind of do a comparison between the 2.

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00:30:16.005 --> 00:30:19.749

Isaiah Osborne: So does he give us the Csv. For.

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00:30:20.090 --> 00:30:25.688

Richard Hoehn: Morgan. So I'm looking at it right now. Here, I'm gonna just share what I'm looking at here.

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00:30:31.550 --> 00:30:34.269

Hector Rogel Jr.: It looks like they gave us a mortgage covenant data.

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00:30:34.530 --> 00:30:41.710

Richard Hoehn: Yeah, yeah, so so I don't know if you guys see what I see here. So

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00:30:41.830 --> 00:30:45.760

Richard Hoehn: so there's definitely a Csv file here.

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00:30:48.280 --> 00:30:51.930

Richard Hoehn: That I can download. Have you? Have we added it to the

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00:30:52.110 --> 00:30:56.260

Richard Hoehn: to the repo. I I sync the repo the other day.

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00:30:56.991 --> 00:30:58.739

Richard Hoehn: I have not added it yet.

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00:30:58.740 --> 00:31:02.479

Richard Hoehn: I'll add it here in this here 8 0, 8 MB.

181

00:31:05.740 --> 00:31:07.850

Isaiah Osborne: That's a nice large Csv.

182

00:31:08.770 --> 00:31:12.110

Richard Hoehn: Yeah, it's pretty big, actually, for Csv, you know what I mean.

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00:31:12.625 --> 00:31:13.140

Isaiah Osborne: Yeah.

184

00:31:13.540 --> 00:31:18.890

Richard Hoehn: Okay, copy data effect.

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00:31:20.830 --> 00:31:23.740

Richard Hoehn: And it's housing bias. Right?

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00:31:24.240 --> 00:31:24.640

Isaiah Osborne: Right.

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00:31:24.640 --> 00:31:26.739

Richard Hoehn: Somebody already put it in there.

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00:31:26.920 --> 00:31:28.639

Richard Hoehn: You must have already done that.

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00:31:29.050 --> 00:31:30.620

Isaiah Osborne: I guess I guess I did.

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00:31:30.620 --> 00:31:32.459

Richard Hoehn: Yeah. Okay. Good job. Thank you.

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00:31:33.040 --> 00:31:35.379

Richard Hoehn: Okay, so it's already there. There's no point in.

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00:31:38.220 --> 00:31:40.629

Richard Hoehn: And you probably also put the

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00:31:41.900 --> 00:31:50.910

Richard Hoehn: you put in code to housing, so we'll still need to do our own ones. But you've kind of got the starter notebook in in the root of the code

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00:31:51.020 --> 00:31:52.040

Richard Hoehn: right.

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00:31:52.810 --> 00:31:53.410

Isaiah Osborne: Yeah.

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00:31:56.590 --> 00:31:57.580

Richard Hoehn: Okay.

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00:31:59.520 --> 00:32:03.374

Richard Hoehn: Oh, can I? Sorry I'm gonna quickly shift gears.

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00:32:04.640 --> 00:32:11.059

Richard Hoehn: So he hasn't update, upgraded, updated, or upgraded our grade yet.

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00:32:11.190 --> 00:32:17.850

Richard Hoehn: Oddly, he sent me an email last week, so not the week.

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00:32:18.210 --> 00:32:21.840

Richard Hoehn: not last class, but the the class before that.

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00:32:22.250 --> 00:32:26.520

Richard Hoehn: And he sent it to me, and he here, I can probably show it

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00:32:27.436 --> 00:32:35.929

Richard Hoehn: like, he said something like, Oh, I get what I did so in his view, it kind of truncates things so almost.

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00:32:35.930 --> 00:32:36.620

Isaiah Osborne: No.

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00:32:36.620 --> 00:32:38.920

Richard Hoehn: Old repo and our new stuff.

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00:32:39.300 --> 00:32:40.010

Isaiah Osborne: That makes sense.

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00:32:40.010 --> 00:32:43.149

Richard Hoehn: And so that's why he did half right and half wrong

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00:32:44.196 --> 00:32:49.849

Richard Hoehn: respond to email. Because I assumed, okay, you're gonna just take care of it. Now.

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00:32:51.200 --> 00:32:51.740

Isaiah Osborne: Like.

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00:32:52.050 --> 00:32:58.969

Richard Hoehn: I'm gonna wait till Monday, and then I'll just reply to him and say, Hey, you know.

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00:33:00.770 --> 00:33:08.009

Richard Hoehn: bump up our great kind of what what's going on a little bit. And I'll copy you guys on the email again.

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00:33:08.871 --> 00:33:19.719

Richard Hoehn: Just just that. We we don't let that slide because we did the work. I think we should get rewarded for the work.

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00:33:19.720 --> 00:33:20.310

Richard Hoehn: Some credit.

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00:33:20.900 --> 00:33:32.749

Richard Hoehn: Yeah. Yeah. Another another great up, because I mean, we. We pulled a a low 80, I think on that, and and we we did much better than an 80. So

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00:33:34.790 --> 00:33:36.190

Isaiah Osborne: I mean our collaborating right with.

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00:33:36.190 --> 00:33:44.050

Isaiah Osborne: you know. Fill the tank now, and coast towards the end of the semester. Personally, definitely.

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00:33:50.130 --> 00:33:50.740

Richard Hoehn: Yeah.

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00:33:50.740 --> 00:33:53.890

Isaiah Osborne: So the good news is, is looking at this notebook.

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00:33:55.307 --> 00:33:56.970

Isaiah Osborne: It looks like it should be.

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00:33:57.430 --> 00:34:00.259

Isaiah Osborne: I'm I'm I'd say, fairly easy

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00:34:00.936 --> 00:34:05.999

Isaiah Osborne: not that hard to extend it for the new county that we need to use instead.

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00:34:06.700 --> 00:34:07.620

Richard Hoehn: Okay. Hmm.

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00:34:07.620 --> 00:34:14.609

Isaiah Osborne: So thankfully. That doesn't look like it's gonna be too bad. I've actually worked with Geo pandas a little bit. So hopefully, we can wrangle that

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00:34:15.650 --> 00:34:17.270

Isaiah Osborne: data well enough.

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00:34:18.199 --> 00:34:20.909

Richard Hoehn: Okay, is Geo. Pans essentially a

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00:34:21.069 --> 00:34:29.119

Richard Hoehn: a Gis kind of way of of mapping. Okay, yeah.

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00:34:33.790 --> 00:34:37.850

Isaiah Osborne: So we should be able to get that to work hopefully.

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00:34:41.510 --> 00:34:52.700

Richard Hoehn: Hey, look! I I don't think we're the the worst group kind of thing like, so I have a I have an evening class with with a bunch of the other guys like

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00:34:53.182 --> 00:35:17.330

Richard Hoehn: like, I've got this evening class called data Data Science Seminar, or something like that. And there's 4 of us in this class 4 with 4 students and a professor, and 3 of them are also in our ethics class kind of thing at different groups. And they're like, you know, we're we're not doing great kind of thing. So.

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00:35:18.570 --> 00:35:21.800

Isaiah Osborne: Y'all doing in your seminar class, because I have to take that next semester.

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00:35:22.458 --> 00:35:25.729

Richard Hoehn: You know it's kind of hard to say.

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00:35:26.400 --> 00:35:28.729

Isaiah Osborne: That seems weird, saying that.

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00:35:30.070 --> 00:35:31.000

Richard Hoehn: This is.

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00:35:31.000 --> 00:35:31.990

Karson Woods: 90.

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00:35:33.218 --> 00:35:36.641

Richard Hoehn: Yeah, yeah, I'm not in love with it.

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00:35:37.910 --> 00:35:44.479

Richard Hoehn: the professor is taking it pretty easy. She's very new. She just got the Mtsu, like last semester.

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00:35:46.850 --> 00:35:57.400

Richard Hoehn: we we spend a lot of time on mathematical approaches to gradient descent and stuff like that matrix math

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00:35:58.650 --> 00:36:08.510

Richard Hoehn: and then she's going through convolutional neural networks, recurrent neural networks and attention mechanisms or transformers.

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00:36:08.680 --> 00:36:09.170

Isaiah Osborne: Oh, nice!

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00:36:11.000 --> 00:36:15.178

Richard Hoehn: And and so it's nice to have a recap. To be quite honest.

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00:36:18.090 --> 00:36:25.650

Richard Hoehn: we I just got my midterm like like she just posts it. She shouldn't have class last week.

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00:36:26.090 --> 00:36:27.790

Richard Hoehn: It's it's

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00:36:28.350 --> 00:36:38.889

Richard Hoehn: it's much easier than this class or my statistics class. To be quite honest, it's easier. She does like you showing up.

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00:36:42.020 --> 00:36:52.930

Richard Hoehn: and she likes you participate. I mean, there's 4 students like, I've never had a class with only 4 students, right? So like spotlight kind of like you have to interact.

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00:36:53.493 --> 00:37:06.880

Richard Hoehn: So that's good and bad, because you know how sometimes you just don't even know what they're talking about and class. You just can't not understand that kind of thing. So good and bad.

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00:37:07.160 --> 00:37:11.189

Isaiah Osborne: Yeah, 4 people is crazy. The smallest I think I've

ever been. It was like 8.

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00:37:11.560 --> 00:37:20.750

Richard Hoehn: Yeah, no, it's 4. We're we're all just sitting next to each other, is it's kind of interesting, like, it's a big classroom, you know.

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00:37:21.200 --> 00:37:28.520

Richard Hoehn: you know, like 2 screens kind of thing, one of those classrooms. And we're just seeing. And she's kind of in front of it like

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00:37:28.680 --> 00:37:31.579

Richard Hoehn: it's it's okay. I wish

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00:37:32.840 --> 00:37:42.190

Richard Hoehn: I wish I would have had a different class like that, like a you know, deep, deep learning, or Nlp, or some, you know, some

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00:37:42.690 --> 00:37:47.600

Richard Hoehn: heavy kind of computer sciencey class. This one

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00:37:49.432 --> 00:37:52.530

Richard Hoehn: having a small class light doesn't really

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00:37:52.800 --> 00:37:59.230

Richard Hoehn: isn't doesn't really translate into, you know, getting a lot of out of it from a professor, in my opinion.

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00:37:59.400 --> 00:38:01.880

Richard Hoehn: right? Like the I mean

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00:38:02.050 --> 00:38:08.159

Richard Hoehn: it would. I don't know if you guys there, there's this really cool class called Deep Learning, with Dr. Phillips.

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00:38:08.920 --> 00:38:34.019

Richard Hoehn: If you can take it, you should, and it's very difficult, and he makes it work. It's it's super. But I thought it was. Really, he's a really good teacher, right? And like, it's really

interesting like for me. It started clicking the whole, you know, parameters and weights and neural networks and attention, and all that kind of stuff. So.

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00:38:34.440 --> 00:38:37.929

Isaiah Osborne: Yeah, I had him for intro to AI.

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00:38:38.710 --> 00:38:43.059

Richard Hoehn: Oh, like 1,500! Was it 1530 or something?

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00:38:43.060 --> 00:38:44.060

Isaiah Osborne: 50, something like that.

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00:38:44.595 --> 00:38:45.130

Richard Hoehn: Okay.

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00:38:45.661 --> 00:38:48.820

Isaiah Osborne: That is honestly, probably the hardest class I've taken.

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00:38:49.290 --> 00:38:52.309

Richard Hoehn: Oh, yeah, it's not easy with him, but like.

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00:38:52.610 --> 00:38:53.080

Isaiah Osborne: He's great.

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00:38:53.080 --> 00:38:56.310

Richard Hoehn: It was interesting, though kind of thing.

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00:38:56.500 --> 00:38:57.310

Isaiah Osborne: Yeah.

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00:38:57.720 --> 00:38:58.480

Richard Hoehn: Yeah.

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00:39:01.500 --> 00:39:03.925

Richard Hoehn: yeah. I mean, he's he's pretty.

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00:39:06.630 --> 00:39:15.495

Richard Hoehn: yeah, he, he puts a lot of pressure on you, right? Like, you had to write a paper, and you know the tests are not easy, and the and all that. But

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00:39:17.250 --> 00:39:21.289

Richard Hoehn: I guess sometimes I feel like sometimes these classes at least I'm

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00:39:21.910 --> 00:39:25.080

Richard Hoehn: push to learn kind of thing, whereas.

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00:39:25.080 --> 00:39:25.399

Isaiah Osborne: Oh, yeah.

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00:39:25.400 --> 00:39:31.490

Richard Hoehn: Plus. I'm just like, Okay, I'm I'm just waiting for waiting for it to be done so.

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00:39:32.270 --> 00:39:37.319

Richard Hoehn: and that might be a a little bit of an age thing right? Like, you know, like

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00:39:37.990 --> 00:39:43.379

Richard Hoehn: I'm I'm already working. I don't need my degree to get them. My, you know.

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00:39:43.840 --> 00:39:44.340

Isaiah Osborne: Yeah.

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00:39:44.340 --> 00:39:56.009

Richard Hoehn: Like. I don't know. My, if you guys want to know my background, I I did all my graduate work at Vanderbilt right? This this doesn't do anything right? Kind of thing, like, I just think it's interesting. Kind of thing. So

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00:39:56.180 --> 00:39:58.060

Richard Hoehn: yeah, so.

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00:39:58.060 --> 00:39:59.329

Isaiah Osborne: What did you do at Vanderbilt?

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00:39:59.330 --> 00:40:08.350

Richard Hoehn: I did business at Vanderbilt. So I did computer engineering at at Mtsu. A master's in computer engineering there.

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00:40:08.930 --> 00:40:19.869

Richard Hoehn: I went and got an Mba at Vanderbilt kind of thing, and then went into into the product management side. So I did children's toys.

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00:40:20.450 --> 00:40:27.370

Richard Hoehn: I did loss prevention and inventory management applications.

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00:40:27.630 --> 00:40:29.301

Richard Hoehn: And then I did my own.

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00:40:29.690 --> 00:40:38.079

Richard Hoehn: I I was a a co-founder of a logistics software company called Freight wise kind of thing.

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00:40:38.370 --> 00:40:40.000

Richard Hoehn: and I sold it

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00:40:40.430 --> 00:40:50.979

Richard Hoehn: in 2022, so I guess 3 years ago now. So I mean, it was this, I started in 2015. I sold it 7 years later

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00:40:51.360 --> 00:40:55.540

Richard Hoehn: and and then

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00:40:56.820 --> 00:41:03.080

Richard Hoehn: started school. And then. Now I'm with this new new job, this Ayrton. So I got the

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00:41:04.530 --> 00:41:13.160

Richard Hoehn: got funding from a company from a from a venture capital company to start a new company called Ayrton, which is

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00:41:15.160 --> 00:41:28.100

Richard Hoehn: route optimization for rural American logistics. Kind of thing. So moving products, herbicides, fertilizer seeds.

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00:41:28.360 --> 00:41:32.930

Isaiah Osborne: All that stuff like that. Tractors that's really cool.

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00:41:33.432 --> 00:41:49.049

Richard Hoehn: Drones. So it's really interesting. You were moving drones around. So they have these big big drones like bigger, like 6 foot wide. They have, like 10 of them on a trailer, they fill them up with.

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00:41:51.780 --> 00:41:55.290

Isaiah Osborne: Herbicides, or whatever you know, like stuff.

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00:41:55.490 --> 00:42:13.510

Richard Hoehn: And and they send these 10 drones out all at once over these fields to spray them where they need to be sprayed, and then they land again. Fill them up and spray. There's some guy with a big trailer full of these drones just driving around rural America, and they spray them.

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00:42:13.790 --> 00:42:17.456

Richard Hoehn: and and the idea is that

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00:42:18.490 --> 00:42:25.429

Richard Hoehn: when it rains it's too wet the earth for the tractors to drive on the fields, but the drones can hit it.

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00:42:25.430 --> 00:42:26.460

Isaiah Osborne: Yeah.

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00:42:26.880 --> 00:42:55.670

Richard Hoehn: And so when it rains, say a week, these farmers have a real problem because they can't be spraying their fields throughout the week, so they have a bubble neck, right? Because they only have so many tractors and spraying rake, as they call them, and so, if they can get these drones out sooner to spray the fields. They alleviate the spraying with their tractors. Kind of thing super like. I don't know anything about

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00:42:55.770 --> 00:43:08.500

Richard Hoehn: agriculture. By the way, no idea this. It's just abstract kind of thing. But it's it's a really it's kind of interesting, you know, when I start, you know, thinking about what we're actually doing

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00:43:08.850 --> 00:43:15.948

Richard Hoehn: often. So so I'm knee deep. If anybody's interest, I'm knee deep in these

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00:43:16.900 --> 00:43:25.979

Richard Hoehn: for working on algorithms right now for there's a thing called the Sa Traveling Salesman problem.

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00:43:25.980 --> 00:43:26.730

Richard Hoehn: Oh, yeah, yeah.

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00:43:27.420 --> 00:43:32.099

Richard Hoehn: So so I'm I'm I'm working on that because

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00:43:32.450 --> 00:43:37.410

Richard Hoehn: based on the route. You know what's the best route.

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00:43:37.820 --> 00:43:45.000

Richard Hoehn: And this traveling salesman problem is really based on just distance. Right? So that's easy. It's 1 dimension kind of thing.

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00:43:45.490 --> 00:43:51.179

Richard Hoehn: My vision right now. I have not done it yet, but the

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00:43:51.330 --> 00:44:02.389

Richard Hoehn: the intellectual property of the new company is that we have multi dimension route, meaning.

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00:44:02.390 --> 00:44:02.940

Isaiah Osborne: Right.

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00:44:04.050 --> 00:44:06.200

Richard Hoehn: Distance is fine, right?

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00:44:06.330 --> 00:44:11.700

Richard Hoehn: But the distance should really be a function of how heavy the truck is.

309

00:44:11.950 --> 00:44:36.519

Richard Hoehn: So if I'm moving, if I have to go a hundred miles with a hundred pounds, and then go over a hundred miles with 900 pounds. It might be smarter to 1st do the 900 pounds first, st and then go 100 pounds, because the less weight I'm moving the less fuel my truck needs to move right.

310

00:44:36.520 --> 00:44:37.400

Isaiah Osborne: Simpler.

311

00:44:37.954 --> 00:44:41.835

Richard Hoehn: And that's just one thing like time.

312

00:44:42.870 --> 00:44:53.409

Richard Hoehn: there's another dimension is if you drop stuff off, you can pick stuff up. You can start filling your truck up kind of thing. So it's called, pick up and drop off.

313

00:44:53.690 --> 00:45:03.280

Richard Hoehn: And then in agriculture they take these machines that spray the fields.

314

00:45:04.030 --> 00:45:04.420

Isaiah Osborne: Go ahead!

315

00:45:04.420 --> 00:45:22.909

Richard Hoehn: Sometimes they have to change the chemicals in the in the sprayer, and so they it's called a washout, where they put water in and rinse the nozzles that they're spraying the chemicals. And so that's another dimension. How do you try and arrange that that it's optimal.

316

00:45:23.640 --> 00:45:37.380

Richard Hoehn: Did you wash out the least amount versus your distance that you're driving and all that. So based on time kind of thing. So

washing takes half an hour. So if you have to do less washouts, maybe.
Then the distance.

317

00:45:37.570 --> 00:45:40.970

Richard Hoehn: Then it's better to wash out and not.

318

00:45:41.430 --> 00:45:47.859

Richard Hoehn: you know, drive an hour and a half somewhere else.
Conference, anyway, that's what I'm working on right now.

319

00:45:48.320 --> 00:45:49.820

Isaiah Osborne: That sounds really interesting.

320

00:45:50.230 --> 00:45:57.660

Richard Hoehn: Yeah, I I like it. I think it's it's it's pretty pretty
pretty, you know, if you can solve the problem. It's a really cool
kind of thing.

321

00:45:57.790 --> 00:46:00.980

Richard Hoehn: And the thing is it has to be solved quickly, like it's

322

00:46:01.910 --> 00:46:04.770

Richard Hoehn: anybody can solve the problem. It's about how

323

00:46:05.010 --> 00:46:28.735

Richard Hoehn: people want to be able to drag and drop stuff around.
And it's constantly recalculating. And that's what makes it very
difficult is being very performant and and solving those problems. So
it's kind of like real time on on a web page. That that's my my
biggest problem right now. Not necessarily the math. It's the the
performance of the math kind of thing, you know.

324

00:46:29.260 --> 00:46:36.949

Richard Hoehn: working on a bunch of caching mechanisms. So if I've
already solved this problem, can I find

325

00:46:37.220 --> 00:46:46.020

Richard Hoehn: a similar problem? So I'm working on this idea of
predicting. If

326

00:46:46.610 --> 00:46:51.589

Richard Hoehn: because I've already done something similar. Kind of thing that that's my goal right now.

327

00:46:55.510 --> 00:46:57.390

Richard Hoehn: anyway. Sorry. That was a big.

328

00:46:57.860 --> 00:46:59.320

Isaiah Osborne: Big tangent.

329

00:46:59.320 --> 00:47:05.459

Richard Hoehn: We have a tangent kind of thing. It's it's it's on my brain right now. I can't get rid of it. Kind of thing.

330

00:47:06.140 --> 00:47:07.229

Richard Hoehn: Figure it out.

331

00:47:10.110 --> 00:47:11.200

Isaiah Osborne: No, that sounds cool.

332

00:47:13.840 --> 00:47:15.129

Richard Hoehn: And it's all

333

00:47:16.190 --> 00:47:22.650

Richard Hoehn: I mean, if you you guys are interested like Node, js, so Javascript on the back end.

334

00:47:23.170 --> 00:47:30.979

Richard Hoehn: Primarily. And then front end. I'm using typescript view view dot Js with typescript kind of thing.

335

00:47:31.160 --> 00:47:40.470

Richard Hoehn: So it's you know, even though it's Javascript, it's tightly coupled, or you know, in essence.

336

00:47:42.747 --> 00:47:45.070

Richard Hoehn: what do you call it? Like?

337

00:47:45.380 --> 00:47:48.479

Richard Hoehn: Parameterize kind of thing like interface

338

00:47:49.030 --> 00:47:53.439

Richard Hoehn: built meaning every every object is defined, kind of thing.

339

00:47:54.000 --> 00:47:54.680

Isaiah Osborne: Sure.

340

00:47:59.800 --> 00:48:02.489

Richard Hoehn: Type safe, I guess, is the way to say it.

341

00:48:02.970 --> 00:48:03.959

Isaiah Osborne: Oh, yeah. Okay.

342

00:48:04.060 --> 00:48:09.450

Richard Hoehn: Yeah, meaning like, you can't use a string for an integer, or vice versa

343

00:48:10.100 --> 00:48:25.519

Richard Hoehn: and and types. That's the easy one. But like type safe also means the object. Like you have a route object that needs to have these and these parameters or objects sub objects they have to be in that you know. Type, kind of thing.

344

00:48:25.780 --> 00:48:26.130

Isaiah Osborne: Right.

345

00:48:26.130 --> 00:48:30.980

Richard Hoehn: And so when it gets compiled, even though.

346

00:48:31.090 --> 00:48:33.839

Isaiah Osborne: Javascript is interpreted.

347

00:48:34.570 --> 00:48:41.010

Richard Hoehn: The typescript that compiler checks all your code to make sure that you're not trying to

348

00:48:42.710 --> 00:48:43.359

Isaiah Osborne: Do something. You should.

349

00:48:43.360 --> 00:48:52.790

Richard Hoehn: Make a wrong allocation fine, so it should. The ideas it should reduce bugs kind of thing.

350

00:48:56.570 --> 00:48:59.779

Isaiah Osborne: I've written plenty of bugs with strongly typed, languages, like

351

00:49:00.460 --> 00:49:05.071

Isaiah Osborne: yeah, that Darn, malloc and alec

352

00:49:06.960 --> 00:49:10.020

Richard Hoehn: You can do all sorts of crazy stuff on that. Yeah.

353

00:49:10.020 --> 00:49:10.570

Isaiah Osborne: Yeah.

354

00:49:11.110 --> 00:49:13.270

Richard Hoehn: I know it's funny.

355

00:49:18.910 --> 00:49:31.139

Richard Hoehn: Yeah. So I'll try and work on this. I'm so spring break, I think I told you guys you know, I've got kids. So we're gonna go down to the beach to the usual.

356

00:49:31.280 --> 00:49:36.830

Richard Hoehn: We're leaving Sunday. We're driving down to Destin and coming back Thursday and.

357

00:49:37.050 --> 00:49:37.600

Isaiah Osborne: Okay.

358

00:49:37.800 --> 00:49:42.500

Richard Hoehn: And I don't know if I've ever said I've got 5 kids right? So this is fucking zoo kind of.

359

00:49:42.500 --> 00:49:44.030

Isaiah Osborne: Oh, my goodness!

360

00:49:44.640 --> 00:49:48.040

Richard Hoehn: Yeah. So it's it's it's pretty crazy.

361

00:49:48.450 --> 00:49:50.710

Isaiah Osborne: 4. There's 4 kids in my family so like.

362

00:49:50.710 --> 00:49:53.100

Richard Hoehn: Oh, okay. So you know, yeah.

363

00:49:53.100 --> 00:49:54.710

Isaiah Osborne: I lived, that.

364

00:49:54.860 --> 00:49:55.600

Richard Hoehn: You know.

365

00:49:56.090 --> 00:50:01.379

Richard Hoehn: Yeah. 5. There's 4 girls and one boy. The poor guy.

366

00:50:01.770 --> 00:50:02.430

Isaiah Osborne: Wow!

367

00:50:04.090 --> 00:50:09.860

Richard Hoehn: So it's a it's yeah, we.

368

00:50:10.230 --> 00:50:12.270

Richard Hoehn: I mean, we've got a big, you know.

369

00:50:12.400 --> 00:50:17.399

Richard Hoehn: 7 foot round table that we have dinner around, and all the.

370

00:50:17.570 --> 00:50:18.240

Isaiah Osborne: Yep.

371

00:50:19.400 --> 00:50:24.740

Richard Hoehn: So yeah, they keep me busy. Okay.

372

00:50:24.740 --> 00:50:25.399

Isaiah Osborne: They do?

373

00:50:26.930 --> 00:50:27.590

Richard Hoehn: Hmm.

374

00:50:27.820 --> 00:50:29.959

Isaiah Osborne: But yeah, I'll try to.

375

00:50:30.670 --> 00:50:38.099

Isaiah Osborne: maybe sometime next week go ahead and get the data downloaded for Ramsey County. See if we can get that process.

376

00:50:38.100 --> 00:50:38.860

Richard Hoehn: Okay.

377

00:50:39.420 --> 00:50:42.280

Isaiah Osborne: And then we can start doing like our analysis and stuff on that.

378

00:50:42.810 --> 00:50:48.080

Isaiah Osborne: I think we should probably try to run those like side by side, just to make sure that.

379

00:50:48.450 --> 00:50:49.330

Richard Hoehn: Okay.

380

00:50:50.570 --> 00:50:55.050

Isaiah Osborne: So we don't. Wanna we don't wanna do everything for hit a PIN, or whatever the real one is.

381

00:50:55.390 --> 00:51:01.959

Isaiah Osborne: and then figure out like the day before that. Oh, we need to do Ramsey. To like doing them at the same time might be easier.

382

00:51:02.988 --> 00:51:04.099

Richard Hoehn: Like that approach. Yeah.

383

00:51:06.150 --> 00:51:10.580

Karson Woods: Okay, yeah, how we split up the work today, or just wait until next week for that.

384

00:51:12.522 --> 00:51:19.710

Isaiah Osborne: Might want to go ahead. And are we meeting? So we meet next week like spring break week, or do you mean like

385

00:51:19.930 --> 00:51:22.260

Isaiah Osborne: the week they come come back, and it's due.

386

00:51:22.260 --> 00:51:26.089

Karson Woods: Probably spring break, because they'll give us more time to like actually work on.

387

00:51:26.090 --> 00:51:37.599

Richard Hoehn: Yeah, I mean, I I would be fine to meet next week, sometime, like a a Tuesday or a Wednesday kind of thing to have a look at it, and and you know.

388

00:51:37.730 --> 00:51:46.240

Richard Hoehn: if we could maybe give ourselves the mandate to try and really look at this, and and and then talk through it. Then we could

389

00:51:46.720 --> 00:51:52.850

Richard Hoehn: maybe divvy up then right now. I I don't feel like I would know what we're

390

00:51:53.310 --> 00:51:57.159

Richard Hoehn: assigning each other or divvying up. To be quite honest, so.

391

00:51:58.048 --> 00:52:00.120

Karson Woods: Yeah, that makes sense.

392

00:52:02.240 --> 00:52:05.510

Richard Hoehn: I mean, I'm I'm around in the evenings.

393

00:52:06.790 --> 00:52:07.600

Richard Hoehn: Yeah.

394

00:52:07.710 --> 00:52:11.930

Richard Hoehn: Tuesday, Thursday, you know. Tuesday, Wednesday, Thursday. I mean, we could.

395

00:52:14.340 --> 00:52:19.170

Richard Hoehn: Maybe Wednesday would be easy in the middle of the week. I don't know what you guys are doing, but.

396

00:52:19.910 --> 00:52:30.899

Karson Woods: I mean, either works for me really like it doesn't have to be a long meeting, either. Just something short, so you can get applying going then maybe like one last time, for it's due to make sure everything's in order.

397

00:52:31.870 --> 00:52:32.730

Isaiah Osborne: That makes sense.

398

00:52:35.480 --> 00:52:38.120

Isaiah Osborne: Yeah, I could do any of those days except Tuesday.

399

00:52:38.760 --> 00:52:39.570

Richard Hoehn: Okay.

400

00:52:40.620 --> 00:52:42.130

Isaiah Osborne: It's like Wednesday or Thursday are fine.

401

00:52:43.550 --> 00:52:45.829

Hector Rogel Jr.: How many I may have to do any of the days.

402

00:52:50.870 --> 00:52:51.600

Isaiah Osborne: Awesome.

403

00:52:54.190 --> 00:52:57.250

Richard Hoehn: Like, do you wanna do? Wednesday next week?

404

00:52:58.740 --> 00:52:59.480

Richard Hoehn: Okay.

405

00:52:59.850 --> 00:53:00.510

Isaiah Osborne: That's fine!

406

00:53:00.510 --> 00:53:01.590

Karson Woods: Yeah, that works.

407

00:53:02.700 --> 00:53:07.560

Richard Hoehn: What in the evening again, is that good for you guys?

408

00:53:08.130 --> 00:53:09.050

Karson Woods: Yeah.

409

00:53:09.050 --> 00:53:09.840

Hector Rogel Jr.: Yes.

410

00:53:10.160 --> 00:53:15.759

Richard Hoehn: Like 8 o'clock again, or what? What are you guys thinking?

411

00:53:17.950 --> 00:53:19.709

Richard Hoehn: Or far earlier.

412

00:53:20.220 --> 00:53:22.920

Isaiah Osborne: Maybe slightly earlier, earlier, but I don't really care.

413

00:53:24.700 --> 00:53:26.150

Karson Woods: Maybe like 7.

414

00:53:26.380 --> 00:53:27.300

Richard Hoehn: Okay.

415

00:53:28.510 --> 00:53:30.110

Richard Hoehn: What do you think? Hector?

416

00:53:30.110 --> 00:53:31.479

Hector Rogel Jr.: Sam works with me.

417

00:53:32.350 --> 00:53:32.910

Richard Hoehn: Right.

418

00:53:34.920 --> 00:53:38.200

Richard Hoehn: Okay, I'll I'll set up the zoom thing for 7.

419

00:53:39.430 --> 00:53:42.679

Richard Hoehn: On Wednesday, the 12.th

420

00:53:43.570 --> 00:53:44.290

Isaiah Osborne: Awesome.

421

00:53:45.490 --> 00:53:50.042

Richard Hoehn: Okay, and so

422

00:53:55.550 --> 00:54:06.290

Richard Hoehn: let's make sure we kind of read really through it and looked at Co, you know, some code and data. So we really talk about then divvying up some work, I guess.

423

00:54:07.510 --> 00:54:11.330

Richard Hoehn: Okay, okay, okay.

424

00:54:11.680 --> 00:54:13.440

Isaiah Osborne: Alright sounds good guys.

425

00:54:13.440 --> 00:54:17.280

Richard Hoehn: Okay, I'll send out the invite again.

426

00:54:17.890 --> 00:54:21.009

Richard Hoehn: Hey, Hector, can you quickly send me your

427

00:54:21.370 --> 00:54:28.479

Richard Hoehn: email address in the chat? Because last time I I kind of put it in the zoom thing.

428

00:54:30.640 --> 00:54:32.200
Richard Hoehn: Where's zoom here?

429
00:54:39.590 --> 00:54:44.970
Richard Hoehn: And then I'll just that way. I don't know if it adds it to your calendar or not, but probably easier

430
00:54:48.350 --> 00:54:49.060
Richard Hoehn: cool.

431
00:55:06.550 --> 00:55:12.760
Richard Hoehn: And I'll upload that transcripts and all that kind of stuff after this. Once it emails it back to us.

432
00:55:12.910 --> 00:55:20.690
Richard Hoehn: I think so. We we've got that under. Okay. Well, let's talk. Wednesday, 7 o'clock.

433
00:55:22.740 --> 00:55:29.313
Richard Hoehn: you know. I think I think that's really proactive of our of us kind of thing. That's

434
00:55:29.800 --> 00:55:31.650
Richard Hoehn: it's pretty good, right?

435
00:55:33.400 --> 00:55:34.830
Isaiah Osborne: Yeah, that sounds good guys.

436
00:55:35.130 --> 00:55:43.745
Richard Hoehn: Okay, well, you guys have a good evening. You guys are probably going out now, kind of thing.

437
00:55:44.470 --> 00:55:44.820
Richard Hoehn: Yeah.

438
00:55:44.820 --> 00:55:47.760
Isaiah Osborne: Have a good time going out in half an hour, so.

439
00:55:47.760 --> 00:55:48.545

Richard Hoehn: Yeah.

440

00:55:51.090 --> 00:55:52.689

Isaiah Osborne: All right. Well, y'all enjoy your break.

441

00:55:53.050 --> 00:55:54.179

Richard Hoehn: Okay, take care. Guys.

442

00:55:54.180 --> 00:55:55.150

Hector Rogel Jr.: Alright, bye.

443

00:55:55.150 --> 00:55:56.020

Karson Woods: Bye.